What the Spiking Heidelberg Digits look like

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Introduction

The gamma-gated-sparsity collection trains the COBANet on static MNIST images that are Poisson-encoded into spikes. Spiking Heidelberg Digits (SHD) is a different animal: the data is *already* spikes. Each sample is a spoken digit passed through a model of the inner ear, so it arrives as a list of events — a spike time and the cochlear channel that fired — with no image and no dense array anywhere. Before training a network on it, this entry just looks at the raw data.

Data

SHD is $N = 8156$ training utterances (a held-out test split adds more), each a spoken digit recorded from several speakers and converted to spikes by a Cramer et al. cochlea model. Every utterance is a set of events $\{(t_k, u_k)\}$, where:

- t_k, the **spike time** of event k, in seconds (utterances run ≈ 0.7–1.0 s);
- $u_k \in \{0, ..., 699\}$, the **cochlear channel** that fired — a place code for frequency, low channel ≈ low pitch, high channel ≈ high pitch;
- so an utterance carries $C = 700$ input channels and a median of ≈ 7600 events (range ≈ 2400–14900), and there are 20 classes: the digits 0–9 spoken in German (labels 0–9) then English (labels 10–19).

Nothing here runs the network — the rasters are drawn straight from the raw HDF5. Binning these events onto the model's integration grid (a $\left(T_{\text{steps}}, 700\right)$ spike tensor at the run's Δt) is the trainer's job, not this entry's.

Results

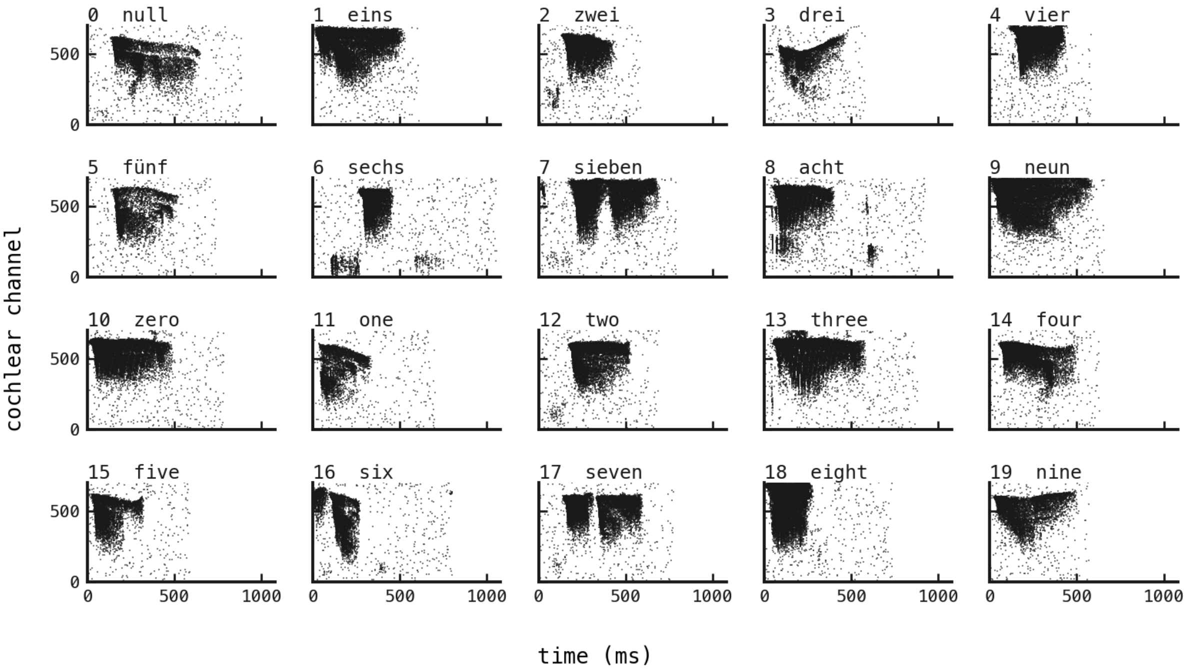

Figure 1: One utterance per class (German 0–9, then English 0–9), each a raster of time (ms) against cochlear channel. **What we expect.** If SHD is learnable, different words should paint different time–frequency signatures. **What we see.** They do: a compact high-channel onset for *zwei*, a long two-lobe sweep for *sieben*/*seven*, a low-channel tail on *sechs*. A diffuse haze of isolated events sits under every panel — the cochlea model's spontaneous background firing, which the classifier has to see past.

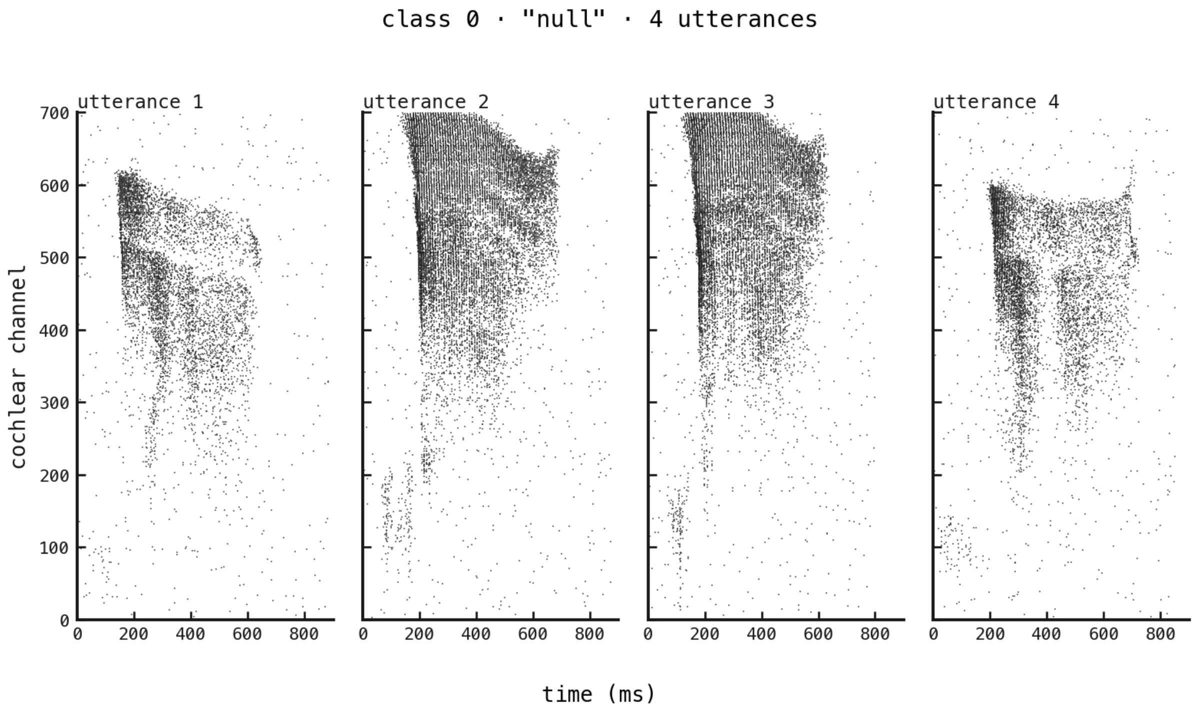

Figure 2: Four different utterances of class 0, *null* — the within-class spread. **What we expect.** Different speakers saying the same word should share a family resemblance while differing in the particulars. **What we see.** Exactly that: all four carry the same broad high-channel onset falling into a mid-channel body, but they differ in duration, event density, and the fine structure of the low-channel tail. This speaker variability is what a classifier trained on SHD must generalise over.

Next steps

The data is well-behaved and visibly class-separable, so it is worth training on. The trainer already accepts it — *tool.py train –dataset shd* bins these events to spikes at the run's Δt and feeds them to the PING network. The next entries take that path: a first PING trained on SHD, then the firing-rate regulariser (Cramer et al.'s $\theta_u = 100$, $s_u = 0.06$) that event-based benchmarks need to keep the hidden rates in check.